Vectors

Chapter Overview

Lines and planes

1. Write vector and Cartesian equations of lines.
2. Describe a plane using two directions or a normal vector.
3. Use scalar products to find angles.

Geometry in three dimensions

4. Find intersections and distinguish skew lines.
5. Calculate shortest distances.
6. Reflect points and lines in planes.

Vector equation of a straight line in 3D

Solution 3

A vector equation describes every point on a straight line.

$$\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Find the points for $\lambda = 0, 1, 2, -1$.

Points

$\lambda = 0: \ (3, 0, -2)$

$\lambda = 1: \ (4, 1, -2)$

$\lambda = 2: \ (5, 2, -2)$

$\lambda = -1: \ (2, -1, -2)$

The first vector locates a fixed point. The second gives the direction.

Changing λ moves along the line in either direction.

The direction vector is non-zero.

$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$

Vector equation of a straight line in 3D

Solution 4

1. Find a line through $(2, 5, 1)$, parallel to direction $(1, 1, 0)$.

1

$$\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

2. Find points on

$$\mathbf{r} = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

at distance 3 from $(3, 4, 4)$.

2

$(\lambda - 4)^2 + (\lambda - 4)^2 + 1 = 9$

$\lambda = 2, 6$

$(1, 2, 3), \quad (5, 6, 3)$

Examples

Solution 5

1. Find a line through $(3, -5, 4)$, parallel to $(7, 0, -3)$.

1

$$\mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 0 \\ -3 \end{pmatrix}$$

2. Find a line through $A(4, 5, -1)$ and $B(6, 3, 2)$.

2

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$$

Use $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. Reversing the direction gives the same line.

Examples

Solution 6

1. The point $(a, b, 0)$ lies on

$$\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + t \begin{pmatrix} 1 \\ -6 \\ -2 \end{pmatrix}$$

Find a, b.

1

$-5 - 2t = 0 \Rightarrow t = -5/2$

$a = 1/2, \qquad b = 17$

2. Show these equations describe the same line:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 8 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

2

The direction vectors are proportional.

At $\lambda = 0$ and $\mu = -2$, both give $(2, 5, -3)$.

Parallel lines with a common point coincide.

Test Your Understanding So Far

Solution 7

Find a vector equation of the line through $A(3, 0, 2)$ and $B(-6, 1, 0)$.

Line

$$\overrightarrow{AB} = \begin{pmatrix} -9 \\ 1 \\ -2 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -9 \\ 1 \\ -2 \end{pmatrix}$$

Examples

Solution 8

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

A circle with centre $P(2, 1, 3)$ meets this line at $A(0, -3, 6)$ and B.

a. Show that P is not on the line.

b. Find B.

a

For $x = 2$, $\lambda = 4$. Then $y = -7 \neq 1$, so P is not on the line.

b

$AP^2 = 4 + 16 + 9 = 29$

$BP^2 = (\lambda - 4)^2 + 4\lambda^2 + (\lambda + 1)^2$

$= 6\lambda^2 - 6\lambda + 17 = 29$

$(\lambda - 2)(\lambda + 1) = 0$

$\lambda = 2$ gives A; hence $\lambda = -1$ gives

$B = (-3, 3, 3)$

Cartesian form of equation of straight line

Solution 9

For a line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$:

$x = a_1 + \lambda b_1,\ y = a_2 + \lambda b_2,\ z = a_3 + \lambda b_3$

Cartesian equation

$$\frac{x - a_1}{b_1} = \frac{y - a_2}{b_2} = \frac{z - a_3}{b_3}$$

This form assumes all three direction components are non-zero. If $b_2 = 0$, write $y = a_2$ separately; do not divide by zero.

A Cartesian equation has no free parameter.

Examples

Solution 10

1. Convert to Cartesian form.

$$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix}$$

1

$$\frac{x - 4}{-1} = \frac{y - 3}{2} = \frac{z + 2}{5}$$

2. Convert to Cartesian form.

$$\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

2

$$x - 2 = \frac{y - 5}{3} = \frac{z}{-2}$$

3. Convert $y = 3x + 2$ to vector form.

3

$$\mathbf{r} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

4. Convert $(x - 2)/3 = y + 5 = z/4$ to vector form.

4

$$\mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$$

5a. The point $(1, p, q)$ lies on

$$\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -4 \\ -9 \end{pmatrix}$$

Find p, q.

a

$\lambda = -1$

$p = 1$

$q = 10$

5b. The point $(1, p, q)$ lies on

$$\mathbf{r} = \begin{pmatrix} -4 \\ 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ -8 \end{pmatrix}$$

Find p, q.

b

$\lambda = 5/2$

$p = -13/2$

$q = -21$

5c. The point $(1, p, q)$ lies on

$$\mathbf{r} = \begin{pmatrix} 16 \\ -9 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

Find p, q.

c

$\lambda = -5$

$p = -19$

$q = -15$

6. Show the lines are parallel:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

$$\frac{x-4}{2} = \frac{y+1}{-4} = \frac{z-3}{-8}$$

6

$$\begin{pmatrix} 2 \\ -4 \\ -8 \end{pmatrix} = -2 \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

The direction vectors are proportional.

10. $P(2, 0, 4)$, $Q(a, 5, 1)$, $R(3, 10, b)$ are collinear. Find a, b.

10

The equal changes in y show that Q is the midpoint of PR.

$$a = \frac{2+3}{2} = \frac{5}{2}$$

$$1 = \frac{4+b}{2} \Rightarrow b = -2$$

11. On

$$\mathbf{r} = \begin{pmatrix} 8 \\ -5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -6 \end{pmatrix}$$

point A has $\lambda = -2$. Find the line through A, parallel to

$$\mathbf{r} = \begin{pmatrix} 10 \\ 3 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$$

11

$A = (2, -7, 16)$

Use the direction of the second line:

$$\mathbf{r} = \begin{pmatrix} 2 \\ -7 \\ 16 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$$

12. The point $A(4, a, 0)$ lies on

$$\mathbf{r} = \begin{pmatrix} 10 \\ 8 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ b \end{pmatrix}$$

a. Find a, b.

b. Find X, where $\lambda = -1$.

a

$10 + \lambda = 4 \Rightarrow \lambda = -6$

$a = 8 - (-6) = 14$

$-12 - 6b = 0 \Rightarrow b = -2$

b

$X = (9, 9, -10)$

13. On

$$\mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

points A, B have parameters 5 and 2. Find AB.

13

$A = (8, 5, -1), \quad B = (5, -1, 5)$

$AB = |5 - 2|\sqrt{1 + 4 + 4} = 9$

14. On

$$\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

points C, A have parameters 4 and 3. A circle with centre C meets the line at A, B. Find $\overrightarrow{OB}$.

14

C is the midpoint of AB, so B has parameter 5.

$$\overrightarrow{OB} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + 5 \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 11 \\ 3 \\ -2 \end{pmatrix}$$

15. A circle has centre $(4, -1, 2)$ and radius $3\sqrt{5}$. It meets

$$x - 5 = \frac{y+1}{3} = \frac{z-6}{-2}$$

at two distinct points. Find them.

Hint: use $(5+\lambda, -1+3\lambda, 6-2\lambda)$ for a point on the line.

15

$(1+\lambda)^2 + (3\lambda)^2 + (4-2\lambda)^2 = 45$

$14\lambda^2 - 14\lambda - 28 = 0$

$\lambda = -1, 2$

$(4, -4, 8), \quad (7, 5, 2)$

16. On

$$\mathbf{r} = \begin{pmatrix} -4 \\ 6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

points A, B have parameters 2 and 5.

a. Find A, B.

b. Find the parallel line through $P(0, 2, 3)$.

c. Points C, D on this second line satisfy $AC = AD = AB$. Show P is their midpoint.

a

$A = (-2, 4, 7), \ B = (1, 1, 10)$

b

$$\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

c

$AB^2 = 27$

$AC^2 = (\lambda+2)^2 + (-\lambda-2)^2 + (\lambda-4)^2$

$= 3\lambda^2 + 24 = 27 \Rightarrow \lambda = \pm 1$

$C = (1, 1, 4), \ D = (-1, 3, 2)$

$\frac{1}{2}(\mathbf{c} + \mathbf{d}) = (0, 2, 3) = \mathbf{p}$

Equation of plane

Solution 19

A plane needs a fixed point and two non-parallel directions.

$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$

$\mathbf{a}$: a point on the plane.

$\mathbf{b}, \mathbf{c}$: independent directions within the plane.

$\lambda, \mu \in \mathbb{R}$: parameters.

For three non-collinear points A, B, C, use $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Examples

Solution 20

1. Find a plane through $A(2, 2, -1), B(3, 2, -1), C(4, 3, 5)$.

1

$$\mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$$

2. Show $P(2, 2, -1)$ lies on

$$\mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

2

$\lambda = -1, \quad \mu = 1$

Substitution gives $(2, 2, -1)$.

Test Your Understanding

Solution 21

[June 2015 Q5] The plane contains $A(1,3,2)$, $B(-1,0,1)$, $C(2,1,0)$. Find its equation in the form $\mathbf{r}\cdot\mathbf{n}=p$.

Plane equation

$\overrightarrow{AB}=(-2,-3,-1), \quad \overrightarrow{AC}=(1,-2,-2)$

A normal is $\mathbf{n}=(4,-5,7)$: its dot product with both directions is zero.

$p=4-15+14=3$

$\mathbf{r}\cdot(4,-5,7)=3$

Cartesian equation of a plane

Solution 22

A normal vector is perpendicular to a line (in 2D) or a plane (in 3D).

A line passes through $(0,3)$ and has normal $(-1,2)$. Find its equation.

Line

$((x,y)-(0,3))\cdot(-1,2)=0$

$-x+2(y-3)=0$

$-x+2y=6$

Cartesian equation of a plane

Solution 23

A plane passes through $(8, 4, -7)$ and has normal $(3, -2, 1)$. Find its equation.

Plane

$3(x-8)-2(y-4)+(z+7)=0$

$3x-2y+z=9$

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Solution 24

3. Find k when the plane contains each point:

$$\mathbf{r}=\begin{pmatrix}2\\-1\\3\end{pmatrix}+\lambda\begin{pmatrix}3\\2\\-2\end{pmatrix}+\mu\begin{pmatrix}1\\-1\\3\end{pmatrix}$$

a. $(7, -1, k)$

a

$3\lambda+\mu=5, \quad 2\lambda-\mu=0$

$\lambda=1,\ \mu=2 \Rightarrow k=7$

b. $(1, k, 11)$

b

$3\lambda+\mu=-1, \quad -2\lambda+3\mu=8$

$\lambda=-1,\ \mu=2 \Rightarrow k=-5$

c. $(k, -4, 10)$

c

$2\lambda-\mu=-3, \quad -2\lambda+3\mu=7$

$\lambda=-1/2,\ \mu=2 \Rightarrow k=5/2$

d. $(10, k, -k)$

d

$2\mu=-2 \Rightarrow \mu=-1$

$3\lambda+\mu=8 \Rightarrow \lambda=3$

$k=6$

7. Show $A(2,2,3)$, $B(1,5,3)$, $C(4,3,-1)$, $D(3,6,-1)$ are coplanar.

8. Show $A(2,3,4)$, $B(2,-1,3)$, $C(5,3,-2)$, $D(-1,-9,8)$ are not coplanar.

7

$\overrightarrow{AB} = (-1,3,0), \quad \overrightarrow{AC} = (2,1,-4)$

$\overrightarrow{AD} = (1,4,-4) = \overrightarrow{AB} + \overrightarrow{AC}$

Thus D belongs to the plane through A, B, C.

8

$\mathbf{r} = (2,3,4) + \lambda(0,-4,-1) + \mu(3,0,-6)$

For D, the first two components force $\mu = -1, \lambda = 3$.

$z = 4 - 3 + 6 = 7 \neq 8$

So D is not in the plane.

9. The plane is

$\mathbf{r} = (3,-2,1) + \lambda(-2,3,5) + \mu(4,2,-3)$

a. Find A at $\lambda = 1, \mu = 2$.

b. Show $B(1,-7,-1)$ is on the plane.

c. Find the line AB.

d. Find $C \neq A$ on this line with $|OC| = |OA|$.

a

$A = (9,5,0)$

b

$\lambda = \mu = -1$ gives $B = (1,-7,-1)$.

c

$$\mathbf{r} = \begin{pmatrix} 9 \\ 5 \\ 0 \end{pmatrix} + t \begin{pmatrix} -8 \\ -12 \\ -1 \end{pmatrix}$$

d

$(9-8t)^2 + (5-12t)^2 + t^2 = 106$

$209t^2 - 264t = 0$

$t = 24/19 \quad (t \neq 0)$

$C = \left(-\frac{21}{19}, -\frac{193}{19}, -\frac{24}{19}\right)$

Challenge

Show the line lies entirely in the plane:

$\Pi : \mathbf{r} = (2, 3, 0) + \lambda(1, -2, 1) + \mu(2, -1, 3)$

$l : \mathbf{r} = (2, 6, 1) + t(5, -7, 6)$

Proof

At $t = 0$, $A = (2, 6, 1)$; it is on the plane at $\lambda = -2, \mu = 1$.

At $t = 1$, $B = (7, -1, 7)$; it is on the plane at $\lambda = 1, \mu = 2$.

A plane contains the whole straight line through two distinct points in it.

The scalar product is the sum of component products:

$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

It is a scalar, not a vector.

1. $(3, 2, 1) \cdot (4, 0, 1)$

1

$12 + 0 + 1 = 13$

2. $(5, -2, 0) \cdot (3, 3, 1)$

2

$15 - 6 + 0 = 9$

3. $(a, 5, 1) \cdot (3, b, 10)$

3

$3a + 5b + 10$

Angle between two vectors

Solution 29

For non-zero vectors:

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

For unit vectors, the scalar product is simply $\cos\theta$.

Find the angle between $(5,3,1)$ and $(1,0,5)$.

Angle

$\mathbf{a}\cdot\mathbf{b} = 10, \quad |\mathbf{a}| = \sqrt{35}, \quad |\mathbf{b}| = \sqrt{26}$

$\theta = \cos^{-1}\dfrac{10}{\sqrt{910}} \approx 70.6^\circ$

Important point...

Solution 30

Choose the correct directions

At B, the angle ABC uses $\overrightarrow{BA}$ and $\overrightarrow{BC}$: both point away from B.

Reversing one vector replaces θ by $180^\circ - \theta$.

Find the angle between $(2,4,-1)$ and $(0,1,8)$.

Angle

$\mathbf{a}\cdot\mathbf{b} = -4$

$\theta = \cos^{-1}\dfrac{-4}{\sqrt{21}\sqrt{65}} \approx 96.2^\circ$

The negative scalar product gives an obtuse angle.

Further Examples

Solution 31

For $A(2,3,5), B(5,0,4), C(4,-3,2)$:

a. Find $\angle ABC$.

b. Find the area of ABC.

a

$\overrightarrow{BA} = (-3,3,1), \quad \overrightarrow{BC} = (-1,-3,-2)$

$\cos B = \dfrac{-8}{\sqrt{19}\sqrt{14}}$

$B \approx 119.4^\circ$

b

$\text{Area} = \dfrac{1}{2}|\overrightarrow{BA}||\overrightarrow{BC}| \sin B$

$= \dfrac{1}{2}\sqrt{266 - 64} = \dfrac{\sqrt{202}}{2} \approx 7.11$

Use the unrounded angle in the area calculation.

Perpendicular vectors

Solution 32

Non-zero vectors are perpendicular exactly when

$\mathbf{a} \cdot \mathbf{b} = 0$

1. Show $(2,3,1)$ and $(1,0,-2)$ are perpendicular.

2. Find a vector perpendicular to both $(-2,5,-4)$ and $(4,-8,5)$.

1

$2(1) + 3(0) + 1(-2) = 0$

2

$-2x + 5y - 4z = 0, \quad 4x - 8y + 5z = 0$

Set $z = 4$, then solve:

$x = 7, \ y = 6$

$\mathbf{n} = (7,6,4)$

Angles between straight lines

Solution 33

$A = (2, 6, -1)$, $B = (3, 4, 1)$. The line l_1 passes through A, B. The line l_2 passes through the origin parallel to $\mathbf{i} + \mathbf{k}$.

Find the acute angle between the lines.

a–b

$$\overrightarrow{AB} = (1, -2, 2)$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$$

c

$$\cos\theta = \frac{|1 + 2|}{3\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

Test Your Understanding

Solution 34

1. Show lines with directions $(2, 1, -1)$ and $(3, -1, 5)$ are perpendicular.

2. $ABCD$ is a parallelogram, with $A(2, -1, 5)$, $B(5, 2, 10)$, $D(-1, 1, 4)$.

   a. Find $\overrightarrow{AB}$.

   b. Give a vector equation of AB.

   c. Find $\angle BAD$.

   d. Find C.

   e. Find the area.

1

$$2(3) + 1(-1) + (-1)(5) = 0$$

2: a–b

$$\overrightarrow{AB} = (3, 3, 5)$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 3 \\ 5 \end{pmatrix}$$

2: angle

$$\overrightarrow{AB} = (3, 3, 5), \quad \overrightarrow{AD} = (-3, 2, -1)$$

$$\cos A = \frac{-8}{\sqrt{43}\sqrt{14}} \Rightarrow A \approx 109.0^\circ$$

2: point and area

$$C = B + D - A = (2, 4, 9)$$

$$\text{Area} = \sqrt{43 \cdot 14 - 64} = \sqrt{538} \approx 23.2$$

12. $A(7,4,4)$, $B(2,2,1)$. Find $\cos\angle AOB$ and the area of triangle OAB.

a

$\mathbf{a}\cdot\mathbf{b}=26, \quad |\mathbf{a}|=9, \quad |\mathbf{b}|=3$

$\cos\angle AOB=\dfrac{26}{27}$

b

$\text{Area}=\dfrac{1}{2}\sqrt{9^2\cdot 3^2-26^2}=\dfrac{\sqrt{53}}{2}$

13. AB is a diameter of a circle centred at the origin. P is another point on the circumference. Prove $AP\perp BP$.

Proof

Write $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=-\mathbf{a}$, $\overrightarrow{OP}=\mathbf{p}$.

$\overrightarrow{AP}\cdot\overrightarrow{BP}=(\mathbf{p}-\mathbf{a})\cdot(\mathbf{p}+\mathbf{a})$

$=|\mathbf{p}|^2-|\mathbf{a}|^2=0$

The two radii have equal length, so the angle is a right angle.

14. $A(5,-1,0)$, $B(2,4,10)$, $C(6,-1,4)$. Find CA, CB, the triangle area, all possible fourth vertices of a parallelogram, and its area.

a

$\overrightarrow{CA} = (-1,0,-4), \quad \overrightarrow{CB} = (-4,5,6)$

b

$\text{Area}(ABC) = \frac{1}{2}\sqrt{17 \cdot 77 - (-20)^2}$

$= \frac{3\sqrt{101}}{2}$

c

$A + C - B = (9,-6,-6)$

$A + B - C = (1,4,6)$

$B + C - A = (3,4,14)$

d

Parallelogram area $= 3\sqrt{101}$

15. $P(1,-1,6)$, $Q(-2,5,4)$, $R(0,3,-5)$. Show PQR is right angled and find the centre and radius of its circumcircle.

Right angle

$\overrightarrow{PQ} = (-3,6,-2), \quad \overrightarrow{QR} = (2,-2,-9)$

$\overrightarrow{PQ} \cdot \overrightarrow{QR} = -6 - 12 + 18 = 0$

Circle

The hypotenuse PR is a diameter.

$\text{Centre} = \left(\frac{1}{2}, 1, \frac{1}{2}\right)$

$\text{Radius} = \frac{|PR|}{2} = \frac{\sqrt{138}}{2}$

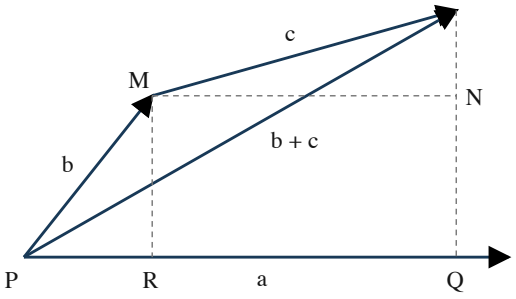
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Solution 39

1. Use the definition of scalar product to prove commutativity.

1

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta = \mathbf{b} \cdot \mathbf{a}$



2a. Show the three projection identities.

2b. Hence prove distributivity.

2a

$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = |\mathbf{a}|\, PQ$

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\, PR$

$\mathbf{a} \cdot \mathbf{c} = |\mathbf{a}|\, RQ$

Each follows by resolving the second vector along $\mathbf{a}$.

2b

$PQ = PR + RQ$

$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = |\mathbf{a}|(PR + RQ)$

$= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$

Vector Equations of Planes

Solution 40

A plane has normal $\mathbf{n}$, and contains a point with position vector $\mathbf{a}$.

For every point $\mathbf{r}$ on the plane, $\mathbf{r} - \mathbf{a}$ is perpendicular to $\mathbf{n}$.

Equation

$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$

$\boxed{\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} = p}$

In Cartesian form, if $\mathbf{n} = (a, b, c)$, the equation is $ax + by + cz = p$.

Example

Solution 41

Find the scalar-product and Cartesian equations of the plane through $(2, 3, -5)$ with normal $(3, 1, -1)$.

Plane

$$p = (2, 3, -5) \cdot (3, 1, -1) = 14$$

$$\mathbf{r} \cdot (3, 1, -1) = 14$$

$$3x + y - z = 14$$

Angle between line and plane

Solution 42

Find the angle between

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 4 \\ -12 \end{pmatrix}$$

and the plane $\mathbf{r} \cdot (2, -2, -1) = 2$.

The line–plane angle is complementary to the acute angle between the line and the normal.

Angle

$$\sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$$

$$= \frac{|6 - 8 + 12|}{13 \cdot 3} = \frac{10}{39}$$

$$\theta \approx 14.9^\circ$$

Test Your Understanding

Solution 43

A plane has equation

$\mathbf{r} = (3,1,2) + \lambda(0,2,-1) + \mu(3,2,2)$

Show $(2,-1,-2)$ is normal to it. Find the acute angle between the plane and the line from $A(1,3,3)$ to $P(3,1,2)$.

a

$(2,-1,-2) \cdot (0,2,-1) = 0$

$(2,-1,-2) \cdot (3,2,2) = 0$

b

$\overrightarrow{AP} = (2,-2,-1)$

$\sin\theta = \dfrac{|4+2+2|}{3 \cdot 3} = \dfrac{8}{9}$

$\theta \approx 62.7^\circ$

Angle between two planes

Solution 44

Find the acute angle between the planes

$\mathbf{r} \cdot (4,4,-7) = 13$

$\mathbf{r} \cdot (7,-4,4) = 6$

Angle

Use their normals and an absolute scalar product:

$\cos\theta = \dfrac{|28-16-28|}{9 \cdot 9} = \dfrac{16}{81}$

$\theta \approx 78.6^\circ$

Test Your Understanding

Find the acute angle between the planes

$\mathbf{r} \cdot (3, -1, 1) = 4$

$\mathbf{r} \cdot (2, 3, 0) = 7$

Solution 45

Angle

$$\cos\theta = \frac{|6-3|}{\sqrt{11}\sqrt{13}}$$

$$\theta \approx 75.5^\circ$$

Exercise 9D - Core Pure Year 1 Pages 187-189

10. Show $A(3, 3, 7)$ lies on both lines and find their acute angle.

$$\frac{x-6}{-1} = \frac{y+3}{2} = \frac{z+2}{3}$$

$$\frac{x+5}{2} = \frac{y-15}{-3} = z - 3$$

Solution 46

a

The parameters are 3 and 4 respectively; each gives $(3, 3, 7)$.

b

$$\cos\theta = \frac{|(-1, 2, 3) \cdot (2, -3, 1)|}{14} = \frac{5}{14}$$

$$\theta = 69.1^\circ$$

11. $\mathbf{r} = \begin{pmatrix}1\\3\\3\end{pmatrix} + \lambda \begin{pmatrix}3\\2\\-1\end{pmatrix}$

$\mathbf{r} = \begin{pmatrix}3\\5\\-2\end{pmatrix} + \mu \begin{pmatrix}-4\\-3\\1\end{pmatrix}$

A is at $\lambda = 3$, B at $\mu = -2$. Find the acute angle between AB and the first line.

Angle

$A = (10, 9, 0)$

$B = (11, 11, -4)$

$AB = (1, 2, -4)$

$\cos\theta = \dfrac{|3 + 4 + 4|}{\sqrt{21}\sqrt{14}} = \dfrac{11}{\sqrt{294}}$

$\theta = 50.1°$

12. $A(3, 5, -1)$, $B(2, -2, 4)$, $C(4, 3, 0)$, $D(1, 4, -3)$. Show the points are not coplanar. Find the acute angle between AD and plane ABC.

a

A normal to ABC is $(1, 2, 3)$, giving $x + 2y + 3z = 10$.

At D, the left side is 0, so D is not in the plane.

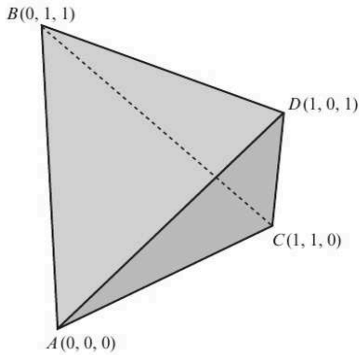
b

$\overrightarrow{AD} = (-2, -1, -2)$

$\sin\theta = \dfrac{|-2 - 2 - 6|}{3\sqrt{14}} = \dfrac{10}{3\sqrt{14}}$

$\theta = 63.0°$

13. A regular tetrahedron has vertices $A(0,0,0)$, $B(0,1,1)$, $C(1,1,0)$, $D(1,0,1)$. Show the angle between adjacent faces is $\cos^{-1}(1/3)$.



Proof

Let E be the midpoint of AD. Both EB and EC are perpendicular to AD.

$E = (1/2, 0, 1/2)$

$EB = (-1/2, 1, 1/2)$

$EC = (1/2, 1, -1/2)$

$$\cos \angle BEC = \frac{1/2}{\sqrt{3/2}\sqrt{3/2}} = \frac{1}{3}$$

Regularity gives the same angle at every edge.

14. Ropes join the top $F(0,0,20)$ of a flagpole to $A(0,8,2)$, $B(12,-5,3)$, $C(-2,6,5)$. The pole is stable only if every pair of ropes makes an angle greater than $15°$. Is it stable?

Decision

$FA = (0, 8, -18), \quad FC = (-2, 6, -15)$

$$\cos\theta = \frac{318}{\sqrt{388}\sqrt{265}}$$

$\theta \approx 7.4° < 15°$

The condition fails for these two ropes, so the pole is not stable.

Points of intersection

Solution 51

Show that these lines intersect and find the intersection:

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 1 \\ 4 \end{pmatrix}$$

Intersection

$3 + \lambda = -5\mu, \quad 1 - 2\lambda = -2 + \mu$

$\lambda = 2, \quad \mu = -1$

Check the third component:

$1 - 2 = 3 - 4 = -1.$

$P = (5, -3, -1)$

Skew lines are non-parallel lines that do not intersect. Always check all three components.

Test Your Understanding So Far

Solution 52

Prove the lines do not meet.

$$\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Proof

Equating x and y:

$\lambda = 2\mu, \quad \lambda = 3 + \mu$

$\mu = 3, \quad \lambda = 6$

The first line has $z = -1$, but the second then has $z = 3$. The equations are inconsistent.

Intersection of line and plane

Solution 53

Find the intersection of

$$\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

and $\mathbf{r} \cdot (1, 2, 3) = 4$.

Intersection

$(-1 + \lambda) + 2(1 + \lambda) + 3(-5 + 2\lambda) = 4$

$9\lambda - 14 = 4 \Rightarrow \lambda = 2$

$P = (1, 3, -1)$

Intersection of lines in Cartesian form

Solution 54

Prove these lines are skew:

$$\frac{x-2}{4} = \frac{y+3}{2} = z - 1$$

$$\frac{x+1}{5} = \frac{y}{4} = \frac{z-4}{-2}$$

Proof

Directions $(4, 2, 1)$ and $(5, 4, -2)$ are not proportional.

Equating x, y gives $\lambda = -9/2, \mu = -3$.

Then $z = -7/2$ on the first line but $z = 10$ on the second.

They do not intersect and are not parallel: they are skew.

2. Show these lines meet and find their intersection.

$$\mathbf{r} = \begin{pmatrix} -6 \\ 0 \\ 11 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ -2 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$$

Intersection

$\lambda - 2\mu = 8, \quad \lambda + \mu = 2$

$\lambda = 4, \quad \mu = -2$

The z coordinates both equal 15.

$P = (-2, -4, 15)$

3. Show the lines do not meet.

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Proof

Equating the first two coordinates gives

$2\lambda - 2\mu = 2, \quad 2\lambda - \mu = 3$

$\lambda = 2, \quad \mu = 1$

But the z coordinates are 4 and -1.

Hence there is no intersection.

7. Find the intersection and the cosine of the acute angle between

$$\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

a

The y coordinates give $\lambda = -1$; then $\mu = 2$. Both lines give

$A = (6, 1, -1)$

b

$$\cos\theta = \frac{|(-1, 1, 2) \cdot (1, 0, -1)|}{\sqrt{6}\sqrt{2}} = \frac{\sqrt{3}}{2}$$

8. Prove these lines are skew.

$$\frac{x}{-3} = \frac{y+1}{5} = \frac{z-2}{4}$$

$$x = \frac{y-1}{-2} = \frac{z+5}{2}$$

Proof

Directions $(-3, 5, 4)$ and $(1, -2, 2)$ are not proportional.

Equating x, y gives $\lambda = -2, \mu = 6$. The corresponding z values are -6 and 7.

They neither meet nor run parallel, so they are skew.

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Solution 59

11. Find where the line meets the plane and their acute angle.

$$\frac{x-3}{5} = \frac{y+2}{3} = 4 - z$$

$$4x + 3y - 2z = -10$$

a

$4(3+5t) + 3(-2+3t) - 2(4-t) = -10$

$31t = -8$

$P = \left(\frac{53}{31}, -\frac{86}{31}, \frac{132}{31}\right)$

b

$\sin\theta = \frac{|(5,3,-1)\cdot(4,3,-2)|}{\sqrt{35}\sqrt{29}} = \frac{31}{\sqrt{1015}}$

$\theta = 76.7°$

Shortest distance between parallel lines

Solution 60

Find the shortest distance between the parallel lines

$$\mathbf{r} = \begin{pmatrix}1\\2\\-1\end{pmatrix} + \lambda\begin{pmatrix}5\\4\\3\end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix}2\\0\\1\end{pmatrix} + \mu\begin{pmatrix}5\\4\\3\end{pmatrix}$$

Distance

Join $A = (1,2,-1)$ to $B = (2+5t, 4t, 1+3t)$.

At the shortest distance, AB is perpendicular to $(5,4,3)$.

$(1+5t, -2+4t, 2+3t)\cdot(5,4,3) = 0$

$3 + 50t = 0 \Rightarrow t = -3/50$

$d^2 = 9 - \frac{9}{50} = \frac{441}{50}$

$d = \frac{21\sqrt{2}}{10}$

Shortest distance between any two lines

Solution 61

Find the shortest distance between

$$\mathbf{r} = \begin{pmatrix}1\\0\\0\end{pmatrix} + \lambda \begin{pmatrix}0\\1\\1\end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix}-1\\3\\-1\end{pmatrix} + \mu \begin{pmatrix}2\\-1\\-1\end{pmatrix}$$

Distance

The shortest joining vector is perpendicular to both directions.

$\overrightarrow{AB} = (-2+2\mu, 3-\mu-\lambda, -1-\mu-\lambda)$

$2-2\mu-2\lambda = 0$

$-6+6\mu+2\lambda = 0$

$\mu = 1, \lambda = 0 \Rightarrow \overrightarrow{AB} = (0,2,-2)$

$d = 2\sqrt{2}$

Shortest distance between point and line

Solution 62

Find the shortest distance from $A(1,2,-1)$ to

$$\frac{x-1}{2} = \frac{y-1}{-2} = \frac{z+3}{-1}$$

and the perpendicular line through A.

Perpendicular

Let the foot be $H = (1+2t, 1-2t, -3-t)$.

$\overrightarrow{AH} = (2t, -1-2t, -2-t)$

$\overrightarrow{AH} \cdot (2,-2,-1) = 9t+4 = 0$

$t = -4/9, \quad AH = \dfrac{\sqrt{29}}{3}$

$\mathbf{r} = (1,2,-1) + s(8,1,14)$

1. Find the shortest distance between the parallel lines

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$$

Distance

Put $t = \mu - \lambda$. A joining vector is $(-2 - 3t, 2 - 4t, 5t)$.

$$d^2 = 8 - 4t + 50t^2 = 50\left(t - \frac{1}{25}\right)^2 + \frac{198}{25}$$

$$d_{\min} = \frac{\sqrt{198}}{5} \approx 2.81$$

2. Find the shortest distance between

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$$

Distance

The shortest joining vector is perpendicular to both lines.

$$AB = \begin{pmatrix} 5 + \lambda + 4\mu \\ -3 - \lambda - 2\mu \\ 1 + \lambda + 3\mu \end{pmatrix}$$

$$\lambda + 3\mu = -3, \quad 9\lambda + 29\mu = -29$$

$$\lambda = 0, \quad \mu = -1$$

$$AB = (1, -1, -2), \quad d = \sqrt{6}$$

4. Find the shortest distance from $(4, 1, -1)$ to

$$\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

Distance

$$d^2 = (1-2t)^2 + (2+t)^2 + (-3+t)^2$$

$$= 6t^2 - 6t + 14 = 6(t - 1/2)^2 + 25/2$$

$$d_{\min} = \frac{5\sqrt{2}}{2} \approx 3.54$$

7. Reflect $P(3, 0, 2)$ in the line

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

Use the perpendicular foot as the midpoint of the point and its reflection.

Reflection

$$PM = (-1+\lambda, -1+2\lambda, 1-\lambda)$$

$$PM \cdot (1, 2, -1) = 6\lambda - 4 = 0$$

$$\lambda = 2/3, \quad PM = (-1/3, 1/3, 1/3)$$

$$Q = P + 2PM = (7/3, 2/3, 8/3)$$

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Solution 67

9. A birdwatcher is at $G(5, 4, 0.7)$ km and can spot birds within 500 m. A kestrel flies in a straight line from $A(3, 5, 0)$ to $B(12, 0, 1.2)$.

Can she spot it? Give a criticism of the model.

Path

$H = (3, 5, 0) + t(9, -5, 1.2), \quad 0 \le t \le 1$

$GH = (-2 + 9t, 1 - 5t, -0.7 + 1.2t)$

a

$GH \cdot (9, -5, 1.2) = 107.44t - 23.84 = 0$

$t = \dfrac{35}{158} \in [0, 1]$

$|GH| \approx 0.45 \text{ km} < 0.5 \text{ km}$

Yes, under this model.

b

A real kestrel is unlikely to fly in a perfectly straight line from A to B.

Exercise 9F - Core Pure Year 1 Pages 191-192

Solution 68

11. Find the shortest distance from $A(3, -1, 2)$ to

$$\frac{x+2}{2} = \frac{y-2}{-1} = \frac{z+1}{-2}$$

and a Cartesian equation of the perpendicular through A.

a

$AM = (-5 + 2t, 3 - t, -3 - 2t)$

$AM \cdot (2, -1, -2) = 9t - 7 = 0$

$t = 7/9, \quad AM = \dfrac{1}{9}(-31, 20, -41)$

$d = \dfrac{13\sqrt{2}}{3}$

b

$$\frac{x-3}{-31} = \frac{y+1}{20} = \frac{z-2}{-41}$$

Shortest Distance from Plane to Origin

Solution 69

For the plane $\mathbf{r} \cdot \mathbf{n} = k$, the shortest distance from the origin is

$$\boxed{d = \frac{|k|}{|\mathbf{n}|}}$$

The perpendicular meets the plane at $H = t\mathbf{n}$.

Derivation

$$t|\mathbf{n}|^2 = k$$

$$OH = |t||\mathbf{n}| = \frac{|k|}{|\mathbf{n}|}$$

The absolute value is essential: distance cannot be negative.

Shortest distance between point and plane

Solution 70

Find the shortest distance from $P(3, 2, -1)$ to $2x - 3y + z = 5$.

Distance

$$d = \frac{|\mathbf{p} \cdot \mathbf{n} - k|}{|\mathbf{n}|}$$

$$= \frac{|6 - 6 - 1 - 5|}{\sqrt{4 + 9 + 1}} = \frac{6}{\sqrt{14}}$$

Use the normal $(2, -3, 1)$ and retain the sign of each coordinate in the numerator.

Test Your Understanding

Solution 71

Find the perpendicular distance from $(6, 2, 12)$ to the plane

$\mathbf{r} \cdot (3, -4, 2) = 5$

Distance

$$d = \frac{|18 - 8 + 24 - 5|}{\sqrt{9 + 16 + 4}}$$

$$= \frac{29}{\sqrt{29}} = \sqrt{29}$$

Reflections of points in planes

Solution 72

Find the distance of $P(1, 3, -2)$ from the plane $\mathbf{r} \cdot (1, 2, 2) = 5$. Then find the reflection Q of P in the plane.

Distance

$$d = \frac{|1 + 6 - 4 - 5|}{3} = \frac{2}{3}$$

Reflection

The foot is $H = P + t(1, 2, 2)$.

$3 + 9t = 5 \Rightarrow t = 2/9$

$$Q = P + 2t(1, 2, 2) = \left(\frac{13}{9}, \frac{35}{9}, -\frac{10}{9}\right)$$

Reflections of lines in planes

Solution 73

Reflect the line

$$\frac{x-2}{2} = \frac{y-4}{-2} = z+6$$

in the plane $2x - 3y + z = 8$.

Fixed point

The line meets the plane when $-14 + 11t = 8$, so $t = 2$ and $A = (6, 0, -4)$.

Reflected line

Reflect its direction $\mathbf{d} = (2, -2, 1)$ using $\mathbf{n} = (2, -3, 1)$:

$$\mathbf{d}' = \mathbf{d} - 2\frac{\mathbf{d}\cdot\mathbf{n}}{|\mathbf{n}|^2}\mathbf{n}$$

$$= \frac{1}{7}(-8, 19, -4)$$

An equivalent direction is $(8, -19, 4)$.

$$\mathbf{r} = (6, 0, -4) + s(8, -19, 4)$$

Exercise 9F (part 2) - Core Pure Year 1 Pages 191-192

Solution 74

6. Find the perpendicular distances to $10x + 10y + 23z = 81$. The normal has length 27.

a. From the origin.

a

$d = 81/27 = 3$

b. From $(-1, -1, 4)$.

b

$$d = \frac{|-10-10+92-81|}{27} = \frac{1}{3}$$

c. From $(2, 1, 3)$.

c

$$d = \frac{|20+10+69-81|}{27} = \frac{2}{3}$$

d. From $(6, 12, -9)$.

d

$$d = \frac{|60+120-207-81|}{27} = 4$$

8. $P = (1, 0, 3)$, and $\Pi : -2x + y + z = 5$. Find the distance from P to Π, then its reflection Q.

a

$$d = \frac{|-2 + 0 + 3 - 5|}{\sqrt{6}} = \frac{2\sqrt{6}}{3}$$

b

The perpendicular foot is

$H = P + t(-2, 1, 1).$

$1 + 6t = 5 \Rightarrow t = 2/3$

$Q = P + 2t(-2, 1, 1)$

$= (-5/3, 4/3, 13/3)$

10. $\Pi_1 : 3x - 2y + 4z = 6$. Find its distance from $(4, -1, 8)$.

For $\Pi_2 : \mathbf{r} = \lambda(2, -2, 3) + \mu(-3, 0, 3)$, show $(2, 5, 2)$ is normal and find the acute angle between the planes.

a

$$d = \frac{|12 + 2 + 32 - 6|}{\sqrt{29}} = \frac{40}{\sqrt{29}}$$

b

$(2, 5, 2) \cdot (2, -2, 3) = 4 - 10 + 6 = 0$

$(2, 5, 2) \cdot (-3, 0, 3) = -6 + 6 = 0$

c

$$\cos\theta = \frac{|6 - 10 + 8|}{\sqrt{29}\sqrt{33}}$$

$\theta = 82.6°$

12. Reflect the line

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -3 \\ 4 \end{pmatrix}$$

in the plane $\mathbf{r} \cdot (1, -1, -1) = 4$.

Fixed point

At the intersection, $1 + 3\lambda = 4$, so $\lambda = 1$ and $A = (6, -4, 6)$.

Reflected line

Reflect $P = (2, -1, 2)$. Its normal projection reaches the plane at parameter $t = 1$, so

$P' = P + 2(1, -1, -1) = (4, -3, 0)$

$A - P' = (2, -1, 6)$

$$\mathbf{r} = \begin{pmatrix} 4 \\ -3 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix}$$